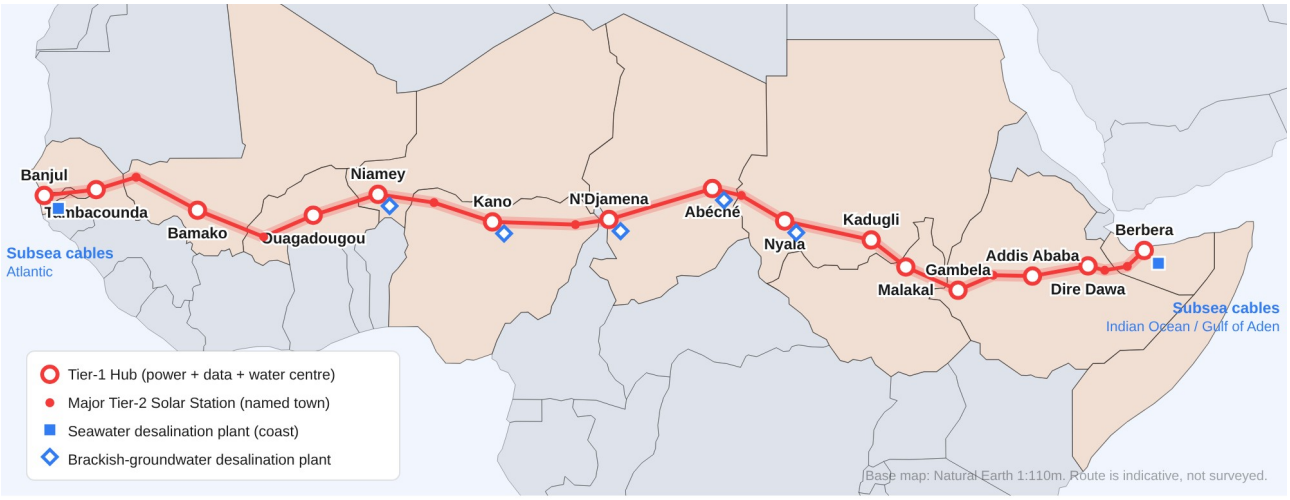
CONCEPT DESIGN

Somalia–Gambia Trans-Sahel Corridor

Water, desalination, solar power and communications along one straight infrastructure strip from Berbera to Banjul



Route	Berbera (Somalia) → Banjul (Gambia), 11 countries
Length	≈ 7,966 km (≈ 4,950 miles) buildable route
Systems	Water main in 6 zones · 7 desalination plants + 30 solar units · ~11 GW solar, 25 GWh batteries · 14 communication layers
Indicative cost	US$44–80 billion over 15–20 years
Status	Concept design. Route, quantities and costs are indicative and not surveyed.
Prepared for	Riad
Date	October 2026

Contents

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1. Executive summary

One straight infrastructure strip from Berbera on Somalia's Gulf of Aden coast to Banjul at the Atlantic end of Gambia. It carries four things side by side: a water main, a solar power grid, a chain of solar stations, and a communication backbone with 14 layers. The communication layer goes in first because it is the cheapest, earns money early, and is needed to monitor everything built after it.

Table 1. Design at a glance

Item	Value
Buildable route length	≈ 7,966 km (≈ 4,950 miles)
Countries crossed	11: Somalia, Ethiopia, South Sudan, Sudan, Chad, Nigeria, Niger, Burkina Faso, Mali, Senegal, Gambia
Stations	~15 Tier-1 hubs, ~100 Tier-2 solar stations, ~400 Tier-3 relay towers
Solar power	~11 GW PV, ~25 GWh batteries, ±500 kV HVDC line (2–3 GW)
Communications	14 layers over 4 independent paths (2 fibre routes, microwave, satellite)
Water	6 hydraulic zones; DN1000–1400 trunk main; reservoirs with 2–5 days buffer
Desalination	2 seawater + 5 brackish plants + 30 solar container units ≈ 790,000 m^3/day
Capital cost	US$44–80 billion, phased over 15–20 years

Distance correction. Somalia to Gambia is 6,692 km (≈ 4,158 miles) in a straight line. A buildable route that follows towns and terrain is about 7,966 km (≈ 4,950 miles). All quantities here are sized for that real length.

Key engineering decision: the water line is split into six zones, not one ocean-to-ocean pipe. Pumping seawater from the coast over the Ethiopian Highlands (≈ 2,355 m) costs at least 6.4 kWh per m^3 for the lift alone, before desalination and friction. Each zone uses its own local sources (rivers, aquifers, seawater desalination at both coasts, and brackish-groundwater desalination at five inland hubs) and connects to the next with valved interties so surplus can move to the driest section.

2. Route and alignment

The line follows the 9–14°N band (the Sahel), which keeps it nearly straight, reaches most regional capitals, and lines up with the existing Great Green Wall programme (a tree-planting belt across the Sahel). Hub spacing averages about 569 km.

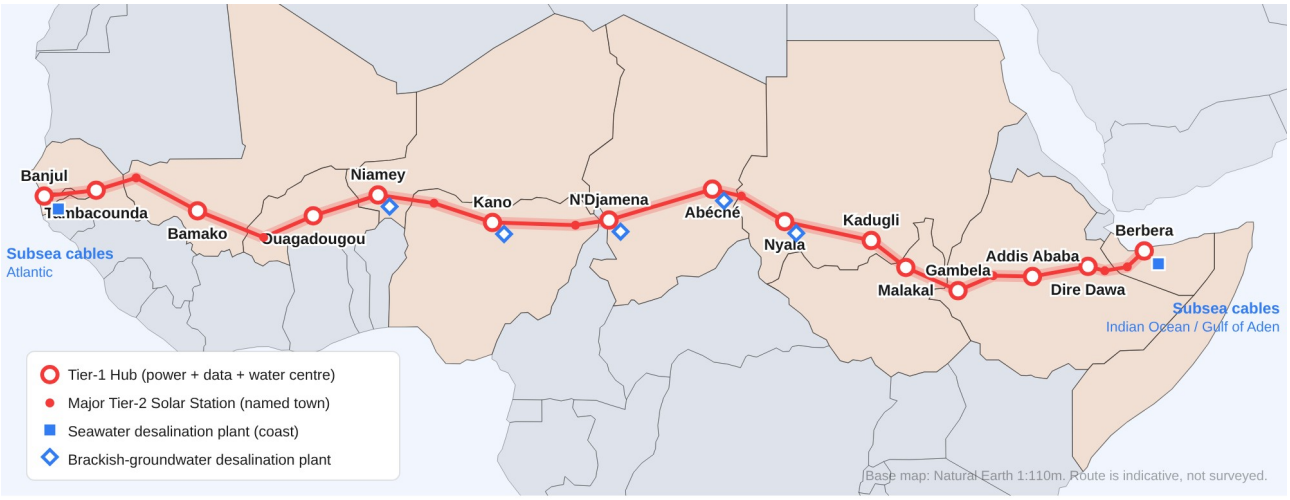


Figure 1. Indicative corridor route with Tier-1 hubs, major Tier-2 stations and desalination plants. Base map: Natural Earth 1:110m.

Table 2. Route segments (lengths include a 12% terrain allowance)

Segment	Country	Length (km)	Cumulative (km)	Elevation at end (m)	Node type
Berbera → Hargeisa	Somalia (Somaliland)	160	160	1,330	Major T2 station
Hargeisa → Jijiga	Somalia (Somaliland) → Ethiopia	157	317	1,610	Major T2 station
Jijiga → Dire Dawa	Ethiopia	119	436	1,200	Tier-1 hub
Dire Dawa → Addis Ababa	Ethiopia	389	825	2,355	Tier-1 hub
Addis Ababa → Nekemte	Ethiopia	271	1,095	2,080	Major T2 station
Nekemte → Gambela	Ethiopia	264	1,359	450	Tier-1 hub
Gambela → Malakal	Ethiopia → South Sudan	393	1,752	390	Tier-1 hub
Malakal → Kadugli	South Sudan → Sudan	301	2,053	500	Tier-1 hub
Kadugli → Nyala	Sudan	605	2,658	670	Tier-1 hub
Nyala → El Geneina	Sudan	343	3,000	800	Major T2 station
El Geneina → Abéché	Sudan → Chad	202	3,202	545	Tier-1 hub
Abéché → N'Djamena	Chad	734	3,936	298	Tier-1 hub
N'Djamena → Maiduguri	Chad → Nigeria	232	4,167	325	Major T2 station
Maiduguri → Kano	Nigeria	566	4,733	480	Tier-1 hub
Kano → Sokoto	Nigeria	420	5,153	270	Major T2 station
Sokoto → Niamey	Nigeria → Niger	383	5,537	210	Tier-1 hub
Niamey → Ouagadougou	Niger → Burkina Faso	463	6,000	305	Tier-1 hub
Ouagadougou → Bobo-Dioulasso	Burkina Faso	370	6,369	432	Major T2 station
Bobo-Dioulasso → Bamako	Burkina Faso → Mali	486	6,856	350	Tier-1 hub
Bamako → Kayes	Mali	472	7,328	50	Major T2 station
Kayes → Tambacounda	Mali → Senegal	283	7,612	50	Tier-1 hub
Tambacounda → Banjul	Senegal → Gambia	354	7,966	5	Tier-1 hub

3. Corridor cross-section

Every layer shares one right-of-way, one service road, and one security perimeter. Building them together saves land, trenching, and patrol cost. The fibre goes in the same trench as the water main.

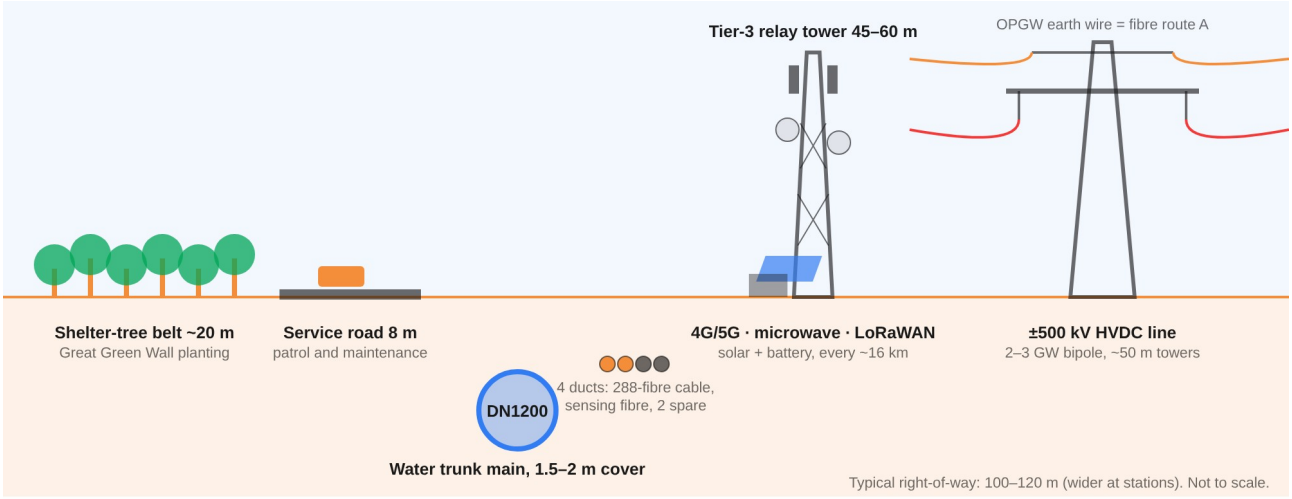


Figure 2. Typical corridor cross-section (not to scale). Right-of-way about 100–120 m, wider at stations.

- **Shelter-tree belt (~20 m).** Great Green Wall planting, watered from the corridor, protects against dust and dunes.
- **Service road (8 m).** Patrol, maintenance and construction access for every layer.
- **Water trunk main.** DN1000–1400 steel or ductile-iron pipe buried with 1.5–2 m cover; four fibre ducts in the same trench.
- **Tier-3 relay towers.** 45–60 m towers every ~16 km, solar and battery powered, carrying 4G/5G, microwave and LoRaWAN.
- **HVDC line.** ±500 kV bipole on ~50 m towers; the earth wire carries the second fibre (OPGW).

4. Solar power and stations

Three tiers of station. Tier-2 spacing (~80 km) matches standard fibre-amplifier spacing and typical pump-station spacing, so one site serves both. Sahel sunshine (~5.5–6.5 kWh/m²/day) gives roughly 1,700–1,900 kWh per kW of panels per year.

Table 3. Station tiers

Tier	Spacing	Count	Solar + storage	What it houses
Tier-1 hub	~500 km, at major cities	15	300 MW–1 GW PV, 0.6–2 GWh battery	HVDC converter, data centre, Internet Exchange Point, regional control room, water treatment or desalination, workshop, airstrip/heliport, security base
Tier-2 solar station	~80 km	~100	20–50 MW PV, 40–100 MWh battery	Pump station and reservoir, fibre amplifier/ROADM, fibre-sensing interrogator, 60 m tower with 5G, depot and drone dock, community water and power
Tier-3 relay site	~16 km (4 between stations)	~400	30–100 kWp PV, 200–400 kWh battery	45–60 m tower: 4G/5G, microwave hop, LoRaWAN gateway, CCTV/thermal camera, pipeline valve chamber, public Wi-Fi

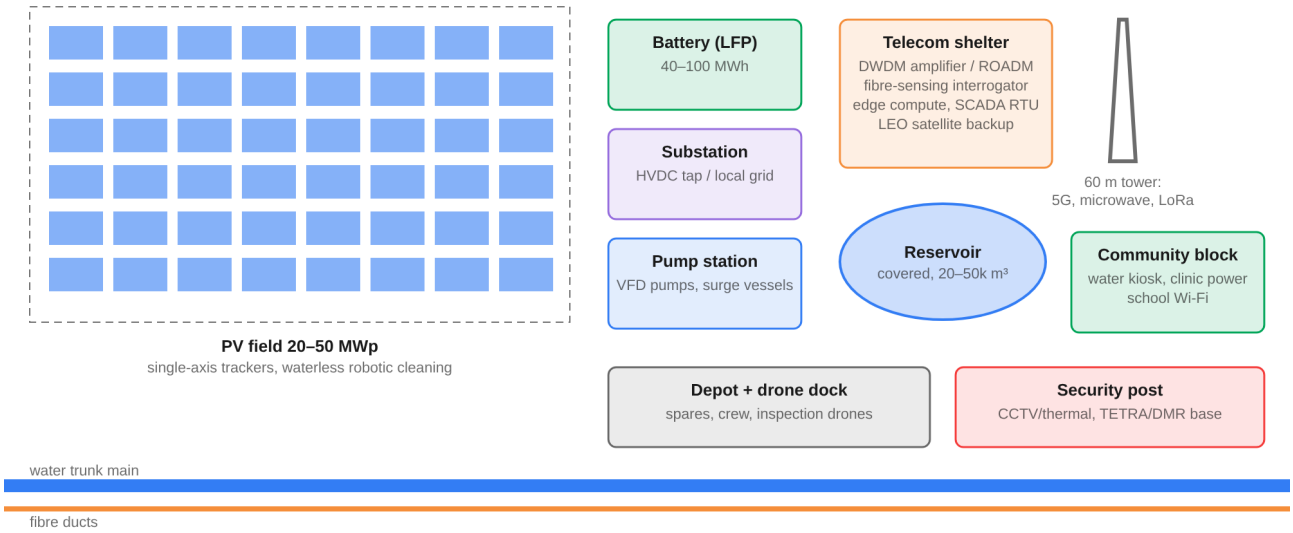


Figure 3. Layout of a typical Tier-2 solar station.

Power system

- **Generation.** About 11 GW of PV in total, clustered at hubs and Tier-2 stations rather than spread along the road edge, because clustered sites are easier to protect and maintain.
- **Storage.** About 25 GWh of lithium iron phosphate (LFP) batteries keep pumps, towers and control systems running overnight.
- **Transmission.** A ±500 kV HVDC bipole (about 2–3 GW) with roughly eight converter stations links the East African and West African power pools and feeds towns along the line.
- **Sahel conditions.** Single-axis trackers, waterless robotic cleaning, and shaded, ventilated equipment shelters for heat above 45 °C.

5. Communications

This covers every practical way to communicate along the full route: fixed, wireless, satellite, sensing, voice, and control. There are four independent transmission paths, so a single cut, outage, or attack does not take the corridor offline.

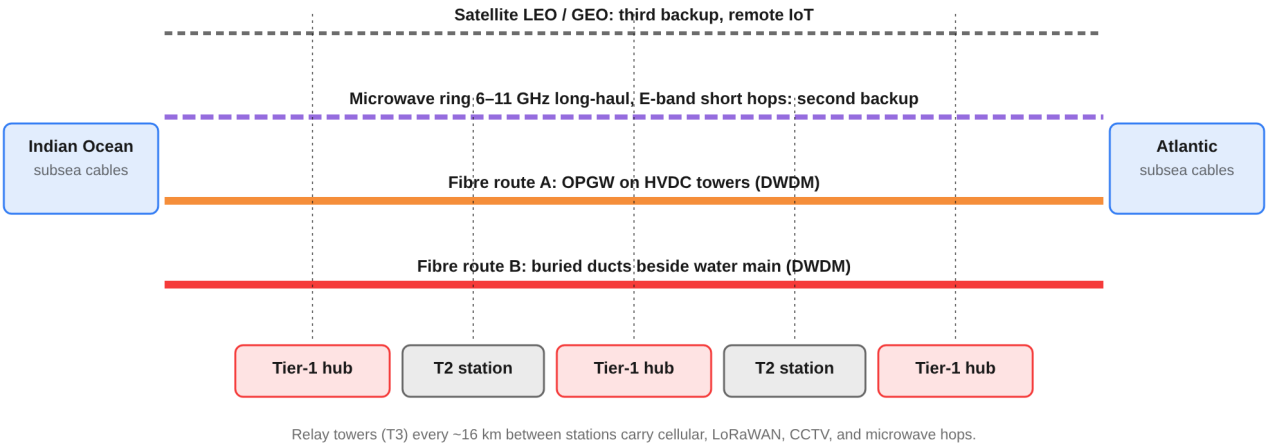


Figure 4. Four independent transmission paths between the Indian Ocean and Atlantic cable landings.

Strategic value. With cable landings at both coasts, the corridor becomes a land route for internet traffic between the Indian Ocean and the Atlantic. That gives Africa, the Gulf, and Asia an alternative to the Red Sea / Suez cable path, where cables have recently been damaged. Selling that capacity can help pay for the

rest of the corridor.

Table 4. The 14 communication layers

Layer	Technology	Where / spacing	Capacity / range	Job on the corridor
1 · Fibre route B (primary)	288-fibre cable in buried HDPE ducts beside the water main; DWDM C+L band	Amplifier every ~80 km (T2), ROADM at every hub	Tens of Tbps per fibre pair with current 400G-per-wavelength optics	Main backbone; wholesale capacity for operators and governments; spare ducts for future cables
2 · Fibre route A (diverse)	OPGW (optical earth wire) on the HVDC towers, 48–96 fibres	Same stations, physically separate path	Same class as route B	Automatic protection switching if route B is cut; power-line teleprotection
3 · Subsea interconnect	Cable landing stations at both ends	East: Berbera / Mogadishu. West: Banjul, with a short spur to Dakar	Several ocean cable systems on each coast (confirm current landing lists)	Turns the corridor into a land bridge between Indian Ocean and Atlantic cables, bypassing the Red Sea / Suez chokepoint
4 · Microwave ring	Licensed 6–11 GHz long-haul hops; 70/80 GHz E-band for short high-capacity hops	Tower to tower, 40–60 km long-haul hops	1–10 Gbps per hop for critical traffic	Second backup; keeps SCADA, voice and security alive during a fibre cut
5 · Public cellular	4G LTE + 5G NR: 700/800 MHz for coverage, 3.5 GHz for capacity in towns; neutral-host shared by national operators	Every T2 and T3 tower	~15–25 km cell radius in flat terrain at low band → a 30–50 km wide coverage strip	Phone and mobile-internet coverage for villages, herders, trucks and travellers; one roaming zone across 11 countries
6 · Private operations network	Private LTE/5G slice for staff, vehicles, cameras and SCADA	Same towers, separate core	Priority and isolation from public traffic	Crew communications, video from drones and cameras, mobile workforce apps
7 · IoT sensor network	LoRaWAN gateways on towers; NB-IoT / LTE-M on cells; satellite IoT where no tower	Every tower; sensors every 1–5 km on the pipe	10–15 km per LoRa gateway; years on one battery per sensor	Pressure, flow, valves, water meters, inverter telemetry, weather, dust, soil moisture for the tree belt
8 · Fibre sensing	Distributed acoustic (DAS) and temperature (DTS) sensing on a dedicated fibre	Interrogator at every T2, ~40 km each direction	Locates events to within metres	Pipeline leak detection, unauthorised digging, vehicle and intrusion alarms
9 · Satellite	LEO broadband terminals at every site; GEO VSAT for broadcast	All tiers	Hundreds of Mbps per LEO terminal	Third backup; construction-phase connectivity; disaster recovery
10 · Mission-critical voice	TETRA or DMR Tier III trunked radio; HF radio as last resort; satellite push-to-talk	Base stations at T2/T1; HF at hubs	Clear voice across the full route	Maintenance, security and emergency crews when everything else is down
11 · Control systems (OT)	IEC 61850 substations, IEC 60870-5-104 / DNP3 telemetry, segregated OT network	All stations, all pumps and valves	Millisecond protection signalling over fibre	Runs the power grid, pumps and valves from the control centres
12 · Community services	Public Wi-Fi at kiosks and schools, community FM radio, telemedicine and e-learning	Every T2 station and nearby villages	Fibre-fed	Direct local benefit; builds community ownership and protection of the line
13 · Drones and aviation	Docked inspection drones flying beyond visual line of sight over private LTE; ADS-B receivers	Drone dock at every T2	Covers each 80 km section daily	Routine inspection of pipe, panels and line; rapid response to sensor alarms
14 · Data and control centres	Data centres + Internet Exchange Points at hubs; Corridor Operations Centre (primary + mirrored backup); Security Operations Centre	Tier-1 hubs	Edge cloud along the route	Keeps local traffic local, monitors all assets, cyber defence

6. Water system

Trunk main of DN1000–1400 steel or ductile-iron pipe. At DN1200 and 1.5 m/s, one line moves ≈ 1.7 m^3/s (≈ 147,000 m^3/day), enough domestic water for roughly 2.9 million people at 50 litres a day. Reservoirs at every station give 2–5 days of buffer.

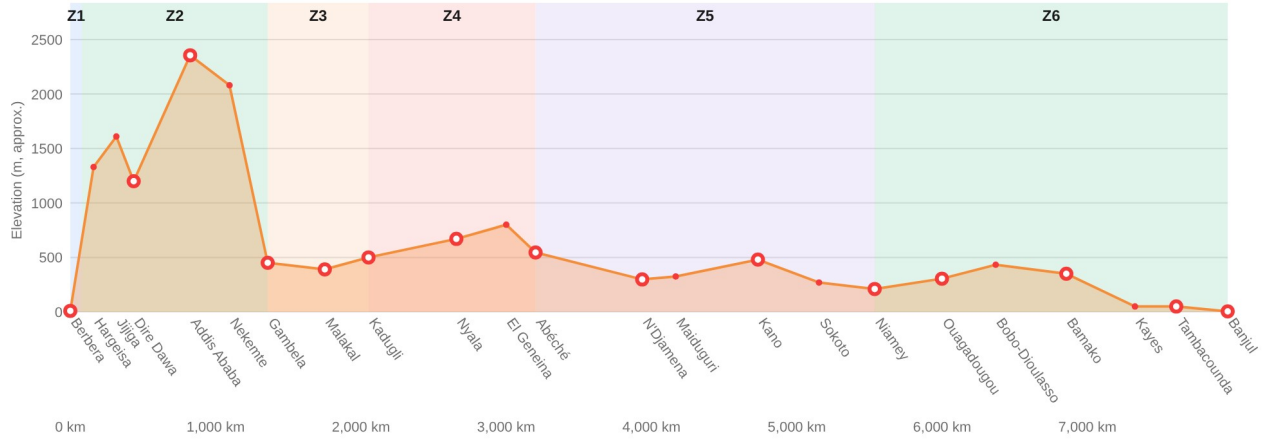


Figure 5. Approximate ground elevation along the route, with the six hydraulic zones.

Table 5. The six hydraulic zones

Zone	Section	Main water sources	Design notes
Z1 Gulf of Aden coast	Berbera → Hargeisa	Solar-powered seawater desalination (reverse osmosis) at Berbera	Steep lift of ~1,300–1,600 m to the plateau; size for towns, not bulk transfer
Z2 Ethiopian Highlands	Jijiga → Nekemte	Rainwater harvesting, small dams, groundwater; heavy rainfall in the west	Highest point (~2,355 m at Addis Ababa). Westward flow is downhill to Gambela, so in-line turbines can recover energy
Z3 Nile basin	Gambela → Malakal	Baro–Sobat / White Nile intakes	Only with Nile Basin agreements; protect wetlands; seasonal-flood intakes
Z4 Darfur–Ouaddaï	Kadugli → Abéché	Deep sedimentary aquifers, wadi sand-dams and recharge basins; brackish-groundwater desalination at Nyala and Abéché; small solar units at remote stations	Driest, most water-stressed section; receives transfers from Z3 and Z5 through interties
Z5 Lake Chad–Niger	N'Djamena → Niamey	Chari–Logone, Lake Chad Basin and Iullemeden aquifers, Niger River at Niamey; brackish-groundwater desalination at N'Djamena, Kano and Niamey	Must not draw down Lake Chad further; managed aquifer recharge in wet season
Z6 Niger–Senegal–Gambia	Ouagadougou → Banjul	Niger River at Bamako, Senegal River (OMVS) at Kayes, Gambia River (OMVG), Atlantic desalination at Banjul	Most water-rich section; can export surplus east to Z5

7. Desalination plants

Three kinds of plant, all powered by the corridor's own solar. Large seawater plants at the two coasts anchor the end zones. Brackish-groundwater plants at five inland hubs turn salty well water into drinking water in the driest zones. Small container units serve remote stations. Total capacity at full build: about 790,000 m^3/day, enough drinking water for about 16 million people at 50 litres/day.

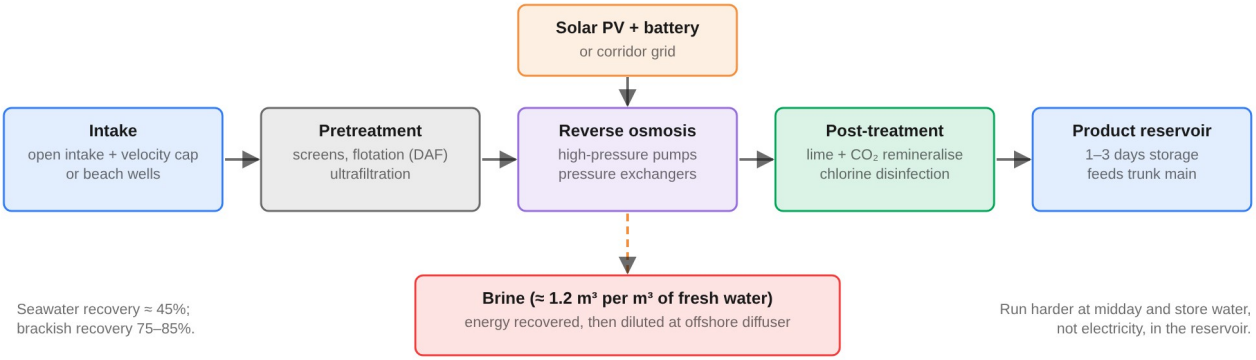


Figure 6. Seawater reverse-osmosis process, powered by corridor solar and batteries.

Table 6. Desalination plants

Plant	Water source	Zone	Capacity m³/day (phase 1 → full)	Power	Brine disposal	Serves
Berbera seawater RO	Gulf of Aden (Indian Ocean), Somalia	Z1	100,000 → 300,000	41 MW avg / ~201 MWp PV	Offshore diffuser 1–2 km out	Port, Berbera, Hargeisa and Jijiga towns; pumped ~1,300–1,600 m up to the plateau
Banjul seawater RO	Atlantic Ocean, Gambia	Z6	150,000 → 300,000	41 MW avg / ~201 MWp PV	Offshore diffuser 1–2 km out	Greater Banjul; pumped east up the Gambia valley to Tambacounda; can back up Dakar by spur
Nyala brackish RO	Sudan (deep wells)	Z4	35,000	1.5 MW avg / ~7 MWp PV	Evaporation ponds + salt recovery	Hub city and nearby stations
Abéché brackish RO	Chad (deep wells)	Z4	35,000	1.5 MW avg / ~7 MWp PV	Evaporation ponds + salt recovery	Hub city and nearby stations
N'Djamena brackish RO	Chad (deep wells)	Z5	50,000	2.1 MW avg / ~10 MWp PV	Evaporation ponds + salt recovery	Hub city and nearby stations
Kano brackish RO	Nigeria (deep wells)	Z5	30,000	1.2 MW avg / ~6 MWp PV	Evaporation ponds + salt recovery	Hub city and nearby stations
Niamey brackish RO	Niger (deep wells)	Z5	25,000	1.0 MW avg / ~5 MWp PV	Evaporation ponds + salt recovery	Hub city and nearby stations
Solar container units (×30)	Remote T2 stations in Z4–Z5	Z4–Z5	100–1,000 each (~15,000 total)	20–150 kWp each, built in	Lined ponds at station	Villages and herders far from the trunk main

Design notes

- **Energy.** Seawater reverse osmosis with pressure exchangers uses about 3–3.5 kWh per m³. Brackish water uses about 0.5–1.5 kWh per m³ because it is much less salty. With solar at roughly 2–4 US cents per kWh, energy costs about 7–14 cents per m³ for seawater.
- **Solar operation.** Plants run at full output in daylight and at reduced output overnight on batteries or the corridor grid. Covered product reservoirs (1–3 days) smooth supply, so less battery is needed.
- **Brine.** At the coasts, brine goes out through a seabed pipe with diffusers 1–2 km offshore, which dilutes it quickly and protects the reefs in the Gulf of Aden. Inland, brine goes to lined evaporation ponds. Salt and minerals are harvested for sale, and the ponds never drain into aquifers or rivers.
- **Water quality.** Red-tide algae in the Gulf of Aden and Atlantic upwelling call for flotation (DAF) plus ultrafiltration before the membranes. Treated water is remineralised so it is healthy to drink and does not corrode the steel trunk main.
- **Cost.** Seawater plants about $1,000–2,000 per m³/day of capacity (≈ $0.6–1.2B for both). Brackish plants about $400–800 per m³/day. Water production cost is roughly $0.5–1.0 per m³ for seawater and $0.2–0.4 per m³ for brackish water. These are covered by the "Desalination" line in the cost chart.
- **Building order.** Berbera and Banjul are built in phase 1 (100,000–150,000 m³/day modules) alongside the pilot segments, then expanded as the trunk main extends inland. Brackish plants follow once well surveys confirm salinity and yield at each hub.

Inland capacities assume brackish groundwater is available at each hub. Hydrogeological surveys must confirm salinity and well yield before design.

8. Cost estimate

Planning-level ranges in 2026 US dollars, built from typical unit costs. Real figures need surveys and bids. Total: $44–80 billion. The water main is about a third of the total, while all 14 communication layers together are about 1–2%.

Table 7. Capital cost range by system, US$ billions

System	Low (US$)	High (US$)
Water mains	$12B	$24B
Desalination	$2B	$4B
Solar PV	$5.5B	$9B
Batteries	$4B	$6B
HVDC	$7B	$12B
Telecom	$0.6B	$1B
Hubs + roads	$4B	$8B
Contingency	$8.8B	$16B
Total	**$44B**	**$80B**

Cost assumptions

- **Water mains.** Includes pump stations, at about $1.5–3 million per km.
- **Solar PV and batteries.** About 11 GW of PV and 25 GWh of storage.
- **HVDC.** The line plus about eight converter stations.
- **Telecom.** Both fibre routes, DWDM equipment, about 500 towers, microwave, satellite, IoT and fibre sensing.
- **Hubs and roads.** Data centres, depots, service road and security.
- **Contingency.** 25% for engineering, land and unknowns.
- **Revenue.** Fibre capacity leasing, power sales, water sales and corridor services can fund operation and maintenance.

9. Implementation phases

Table 8. Phased implementation over about 20 years

Phase	Timing	Scope	Exit criteria
00 · Feasibility, treaties and surveys	Years 0–2	AU / AfDB-backed treaty among 11 states; route survey (LiDAR + satellite); environmental and social impact studies; community consultation; financing structure.	Signed corridor treaty and right-of-way law in each country.
01 · Digital spine first	Years 1–5	Buried fibre, relay towers with solar + battery, cellular, IoT, satellite backup and cable landings. Cheapest layer (about 1–2% of total cost), earns revenue early, and is needed to monitor everything built after it.	Continuous fibre Berbera–Banjul; cellular coverage along the full route.
02 · Solar stations and HVDC	Years 3–10	Build Tier-2 solar stations and Tier-1 hubs; link national grids (East African and West African power pools) with HVDC sections; start with pilot segments Banjul–Tambacounda–Bamako and Berbera–Hargeisa–Jijiga, each with a first desalination module (Banjul and Berbera).	Corridor grid supplies its own pumps, towers and nearby towns.
03 · Water zones	Years 5–18	Build the six hydraulic zones, expand the coastal desalination plants, add the inland brackish-water	Each zone delivers reliable water to towns along its

Phase	Timing	Scope	Exit criteria
		plants, reservoirs and interties; plant the shelter-tree belt using corridor water.	section.
04 · Corridor economy	Year 8 onward	Farming, cold chains, data centres, e-mobility charging and trade hubs along the line; revenue from capacity leasing, power and water sales.	Corridor operations fund their own maintenance.

Communications go first because they cost about 1–2% of the total, earn revenue early and are needed to monitor everything built after them. The first pilot sections are Banjul–Tambacounda–Bamako and Berbera–Hargeisa–Jijiga, each with a first desalination module.

10. Risks and open decisions

Key risks

- **Security.** The route crosses active conflict areas: Somalia, Sudan, the Lake Chad region and the central Sahel. Mitigations include community benefit-sharing at every station (free water kiosk, clinic and school power, local jobs), buried assets, cables with no resale value (fibre, aluminium conductor), fibre-sensing alarms, drones, and building in stable sections first.
- **Shared rivers.** Nile, Niger, Lake Chad, Senegal and Gambia water use requires agreement through the existing basin bodies (Nile Basin Initiative, Niger Basin Authority, Lake Chad Basin Commission, OMVS, OMVG).
- **Climate and terrain.** Dust lowers panel output (waterless robotic cleaning). Heat over 45 °C reduces equipment ratings (shaded, ventilated shelters). Flash floods in dry riverbeds (culverts, deeper pipe crossings). Mobile dunes in the driest stretches (shelter belt, route adjustments).
- **Groundwater.** Inland brackish desalination depends on aquifers that have not yet been surveyed.

Open decisions

1. Eastern start point: Berbera (shortest and straightest; in Somaliland), Mogadishu, or Bosaso.
2. Central route: through northern Nigeria (Kano, shown here) or through Niger (Zinder–Maradi).
3. Governance: a treaty body with one operating company, or national companies under shared standards.
4. Whether to add a continuous solar ribbon along the road edge, or keep solar clustered at stations (recommended, because clustered sites are easier to protect and maintain).

11. Limitations and next steps

This is a concept design. It is intended to describe the idea, test whether it is physically sensible, and size it roughly. It is not a feasibility study or a construction design.

- **Route.** Drawn from city coordinates with a terrain allowance; needs a LiDAR and satellite route survey.
- **Water.** Needs hydrogeological surveys, hydraulic and surge modelling, and basin agreements.
- **Communications.** Current subsea cable landing lists and spectrum licensing must be confirmed country by country.
- **Costs.** Planning-level ranges from typical unit costs; real figures need surveys and bids.
- **Governance.** Needs a treaty among 11 states, right-of-way law in each, and a financing structure (for example AU / AfDB-backed).

Recommended next step: fund a feasibility study and one pilot section at each coast.